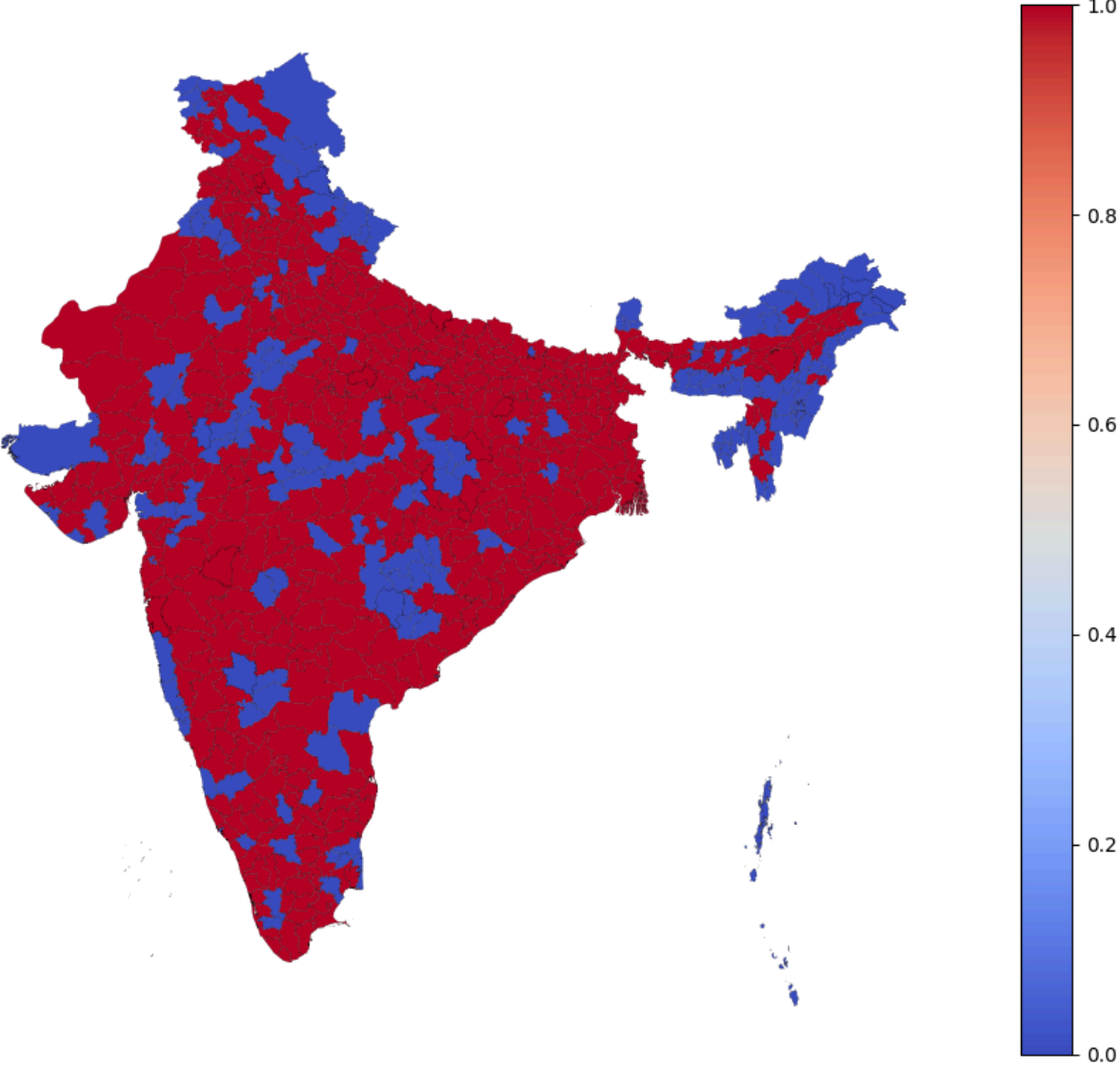
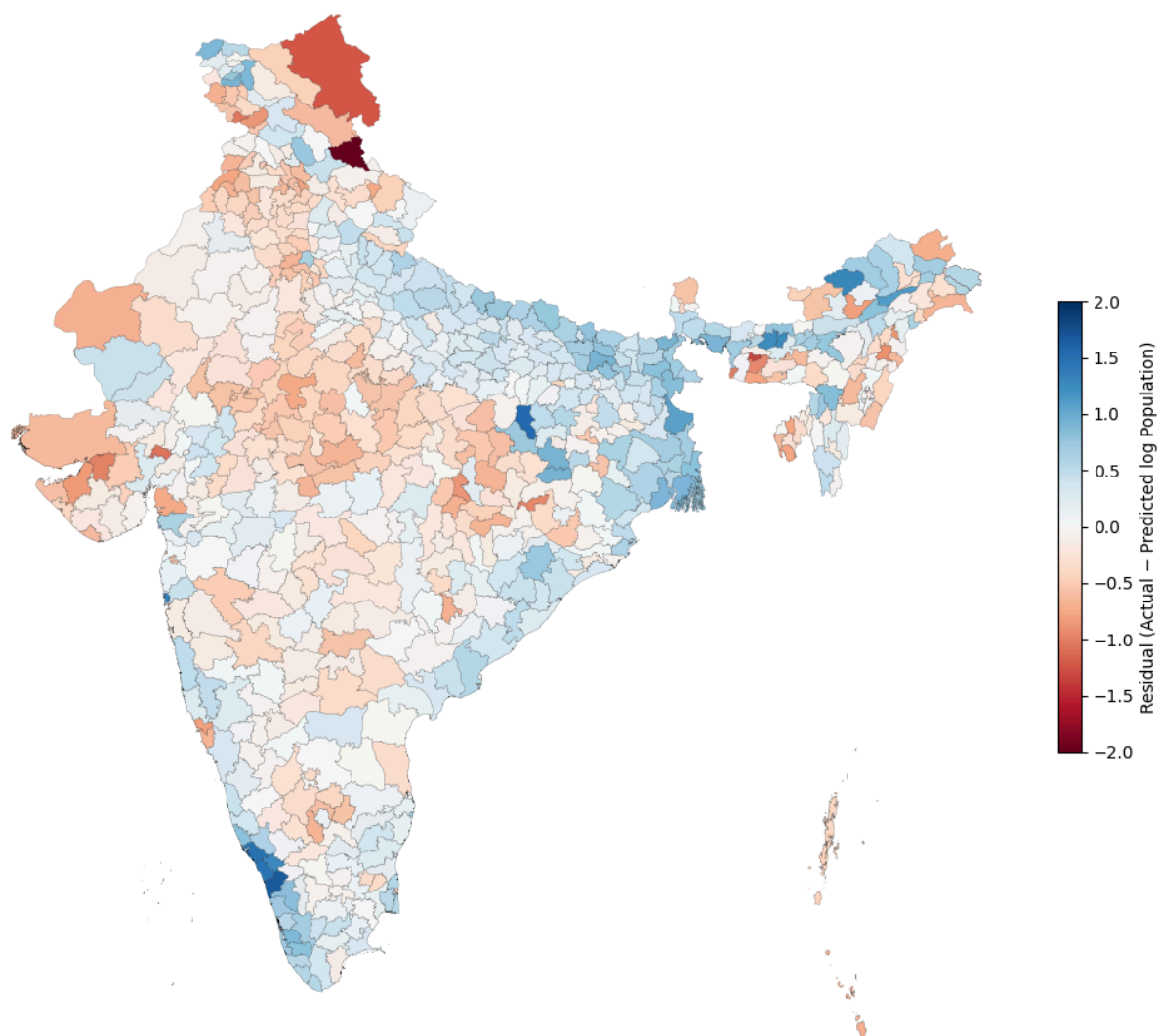


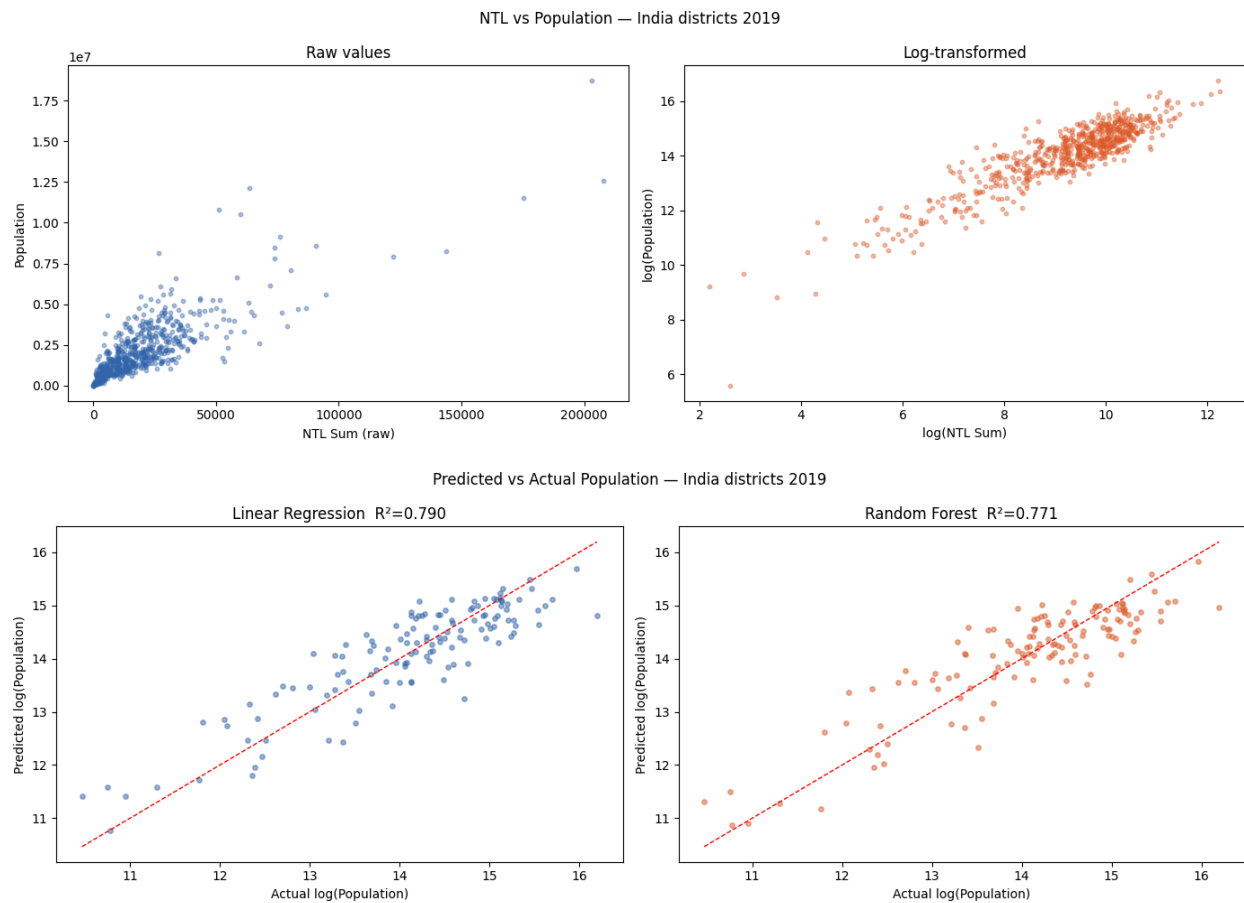
K-Means Urban/Rural Classification — India Districts



NTL Population Model Residuals — India Districts 2019
Red = under-illuminated | Blue = over-illuminated



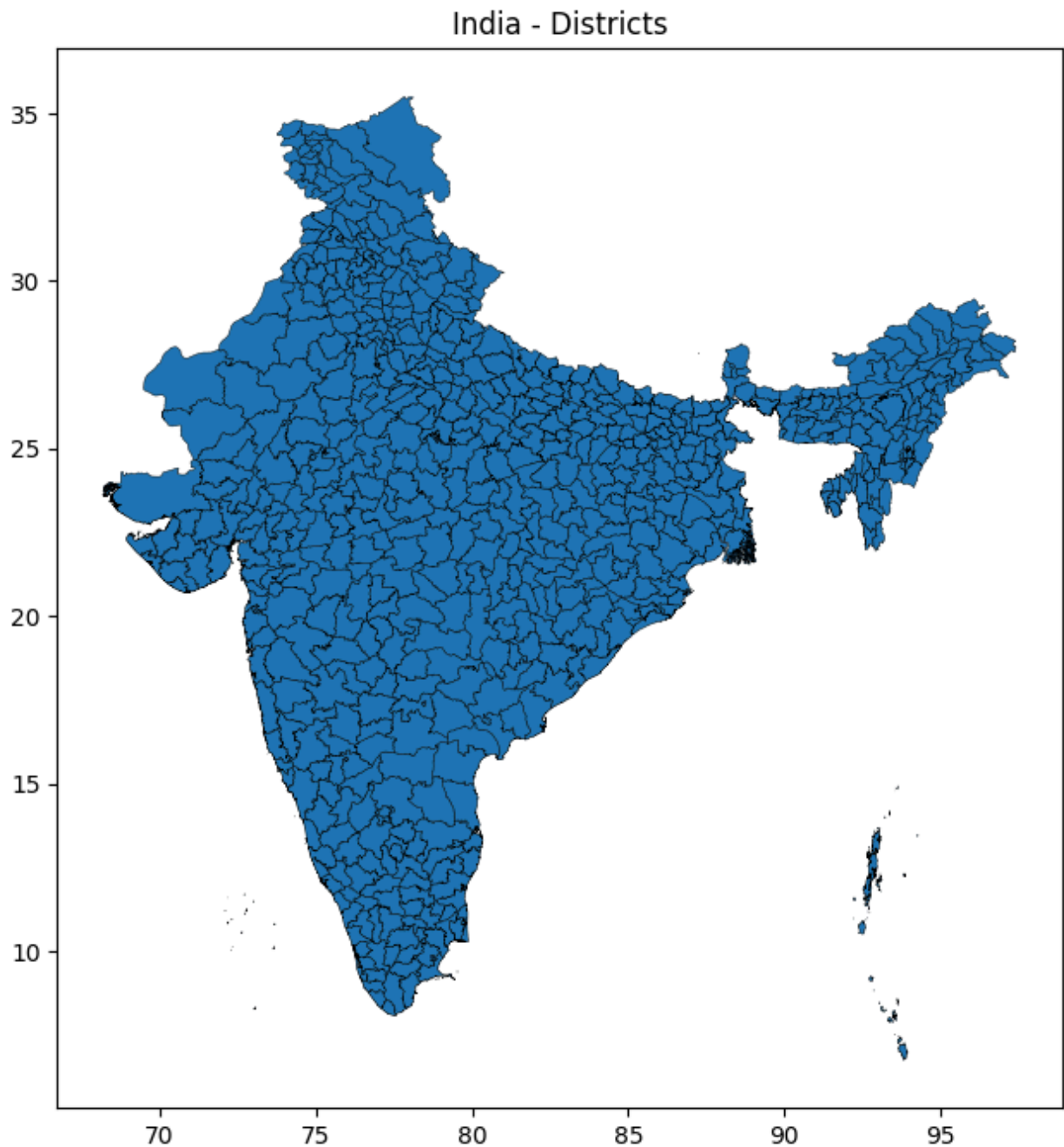
Residual Map Following baseline model fitting, residuals were computed as the difference between actual and predicted log-population for each district and mapped geographically. Red districts indicate systematic under-prediction, where population exceeds what nighttime light would suggest — concentrated in Kerala and Assam, reflecting dense informal settlements and historically low electrification rates. Blue districts indicate systematic over-prediction, where brightness exceeds population expectations — observed in industrial zones of Gujarat, Jharkhand, and the sparsely populated Ladakh region. The spatial clustering of these errors confirmed that model failure was not random but structurally driven by district typology, providing the direct empirical motivation for the urban-rural stratification approach proposed in this study.



NTL vs Population Scatter (Raw + Log) The relationship between nighttime light and population was first examined in raw form, revealing a heavily skewed distribution where the majority of districts compressed near zero while a small number of outliers dominated the scale. This rendered any direct regression meaningless. Upon applying a log transformation to both variables, the relationship resolved into a clean, approximately linear trend across the full range of districts. This transformation forms the core preprocessing step of the methodology and validates the log-linear modelling assumption adopted throughout the study.

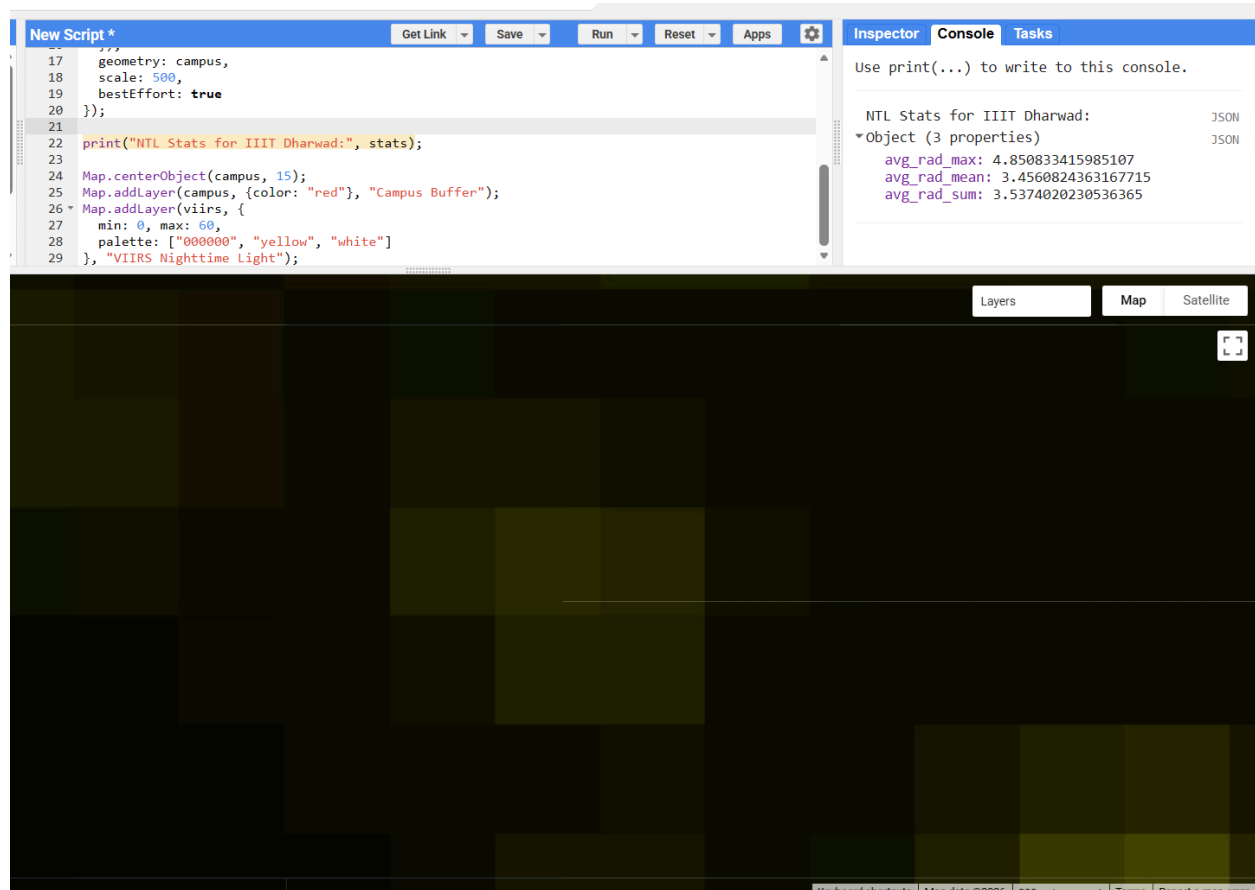
Predicted vs Actual Population The baseline model performance was evaluated by plotting

predicted against actual log-population for all districts. Linear Regression achieved an R^2 of 0.790, with points tracking the diagonal reasonably well across the middle range while showing scatter at the extremes. Random Forest performed slightly worse at $R^2=0.771$, exhibiting characteristic regression-to-the-mean behaviour by clustering predictions around the centre and underperforming at both tails. These results established the performance benchmark against which the proposed stratified approach was subsequently compared



India Districts Map Prior to any modelling, the study area was defined across all 699

administrative districts of India following the post-2014 delimitation. Each district constitutes a single observation in the dataset. The map confirms complete national coverage with no states excluded, including remote territories such as Andaman and Nicobar Islands. This serves as the foundational spatial framework upon which all subsequent analysis was conducted..



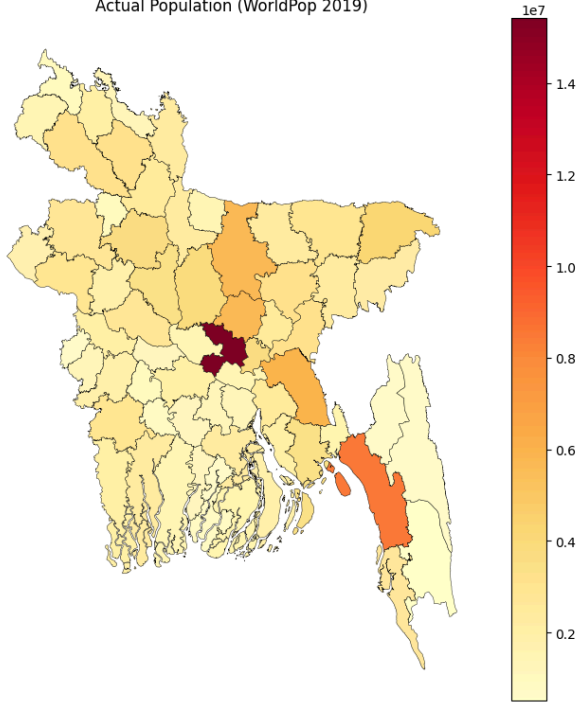
log_ntl_sum: 1.5124
log_ntl_mean: 1.4943
log_ntl_max: 1.7666

Predicted log_pop: 7.6214

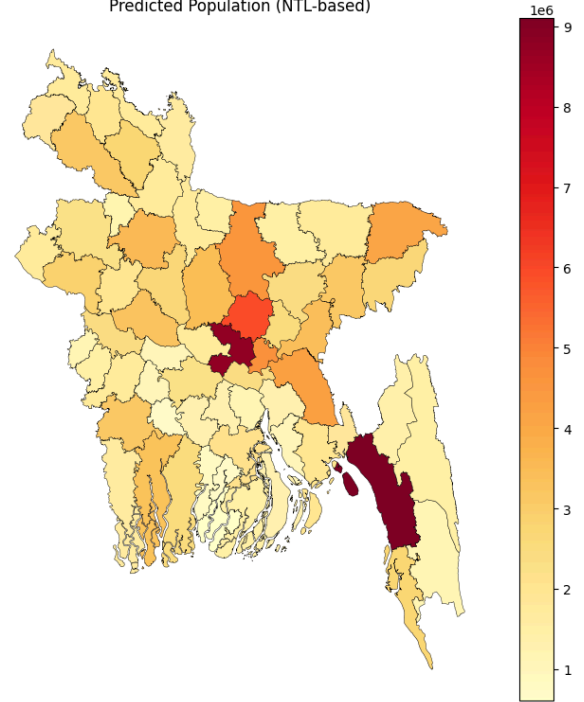
Predicted population: 2,041 (our college campus)

Population Proxy Mapping — Bangladesh 2019

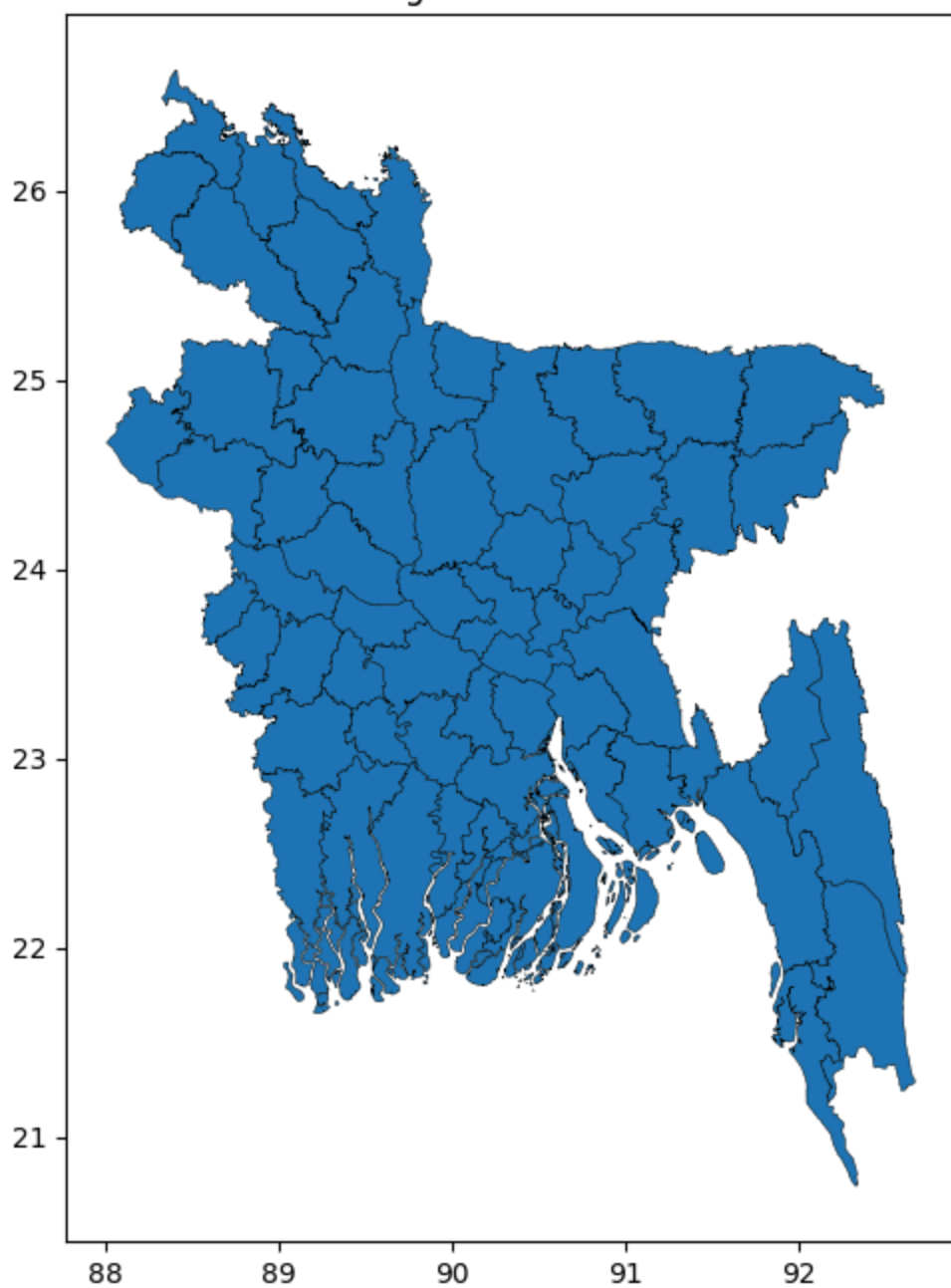
Actual Population (WorldPop 2019)



Predicted Population (NTL-based)



Bangladesh Districts



Residual Map Comparison — India Districts 2019

